220B Homework 4

Due: Sunday, June 30

Problem 1: Pitman's estimator for a Uniform location family (50 pts)

Let $X_1, \dots, X_n$ be iid from the **Uniform location family**

$$f(x \mid \theta) \;=\; \mathbf{1}\{\theta - 1/2 \le x \le \theta + 1/2\}, \qquad \theta \in \mathbb{R}.$$

Estimate θ under squared-error loss $L(\theta, a) = (\theta - a)^2$. This problem is invariant under the location group $\mathcal{G} = \{g_c : g_c(x) = x + c\}$.

(a) **(20)** Show that the joint density of the sample is

$$\prod_{i=1}^{n} f(x_i \mid \theta) \;=\; \mathbf{1}\{x_{(n)} - 1/2 \le \theta \le x_{(1)} + 1/2\},$$

where $x_{(1)} = \min_i x_i$ and $x_{(n)} = \max_i x_i$. Then derive the Pitman estimator (best equivariant estimator under squared-error loss) for θ, and show that

$$\delta_P(X_1, \dots, X_n) \;=\; \frac{X_{(1)} + X_{(n)}}{2}.$$

(b) **(30)** Compute the risk of the midrange in closed form. (Hint: change variables to $Y_i = X_i - \theta \sim \mathrm{Uniform}(-1/2, 1/2)$ and use the joint distribution of $(Y_{(1)}, Y_{(n)})$. For iid $\mathrm{Uniform}(0,1)$ samples, the order statistics satisfy $\mathrm{Var}(U_{(1)}) = \mathrm{Var}(U_{(n)}) = n/[(n+1)^2(n+2)]$ and $\mathrm{Cov}(U_{(1)}, U_{(n)}) = 1/[(n+1)^2(n+2)]$.) Compute the risk of the sample mean $\bar{X}$ as an estimator of θ. Show that the ratio $R(\theta, \delta_P)/R(\theta, \bar{X})$ does not depend on θ, and that it tends to 0 as $n \to \infty$.

Problem 2: SURE for soft thresholding (50 pts)

Let $\mathbf{X} \sim \mathcal{N}_p(\theta, I_p)$ with $p \geq 1$, under squared-error loss. For $\lambda \geq 0$, the **soft-thresholding estimator** is

$$\delta^\lambda(\mathbf{x})_i \;=\; \mathrm{sign}(x_i) \cdot (|x_i| - \lambda)^+ \;=\; \begin{cases} x_i - \lambda & \text{if } x_i > \lambda, \\ 0 & \text{if } |x_i| \leq \lambda, \\ x_i + \lambda & \text{if } x_i < -\lambda. \end{cases}$$

Writing $\delta^\lambda(\mathbf{x}) = \mathbf{x} + \mathbf{g}(\mathbf{x})$, one has $g_i(\mathbf{x}) = -\mathrm{sign}(x_i)\,\min(|x_i|, \lambda)$. (You may take this as given; the non-differentiable set $\bigcup_i \{|x_i| = \lambda\}$ has Lebesgue measure zero.)

(a) **(16)** Compute $\partial g_i / \partial x_i$ at every point where it exists. Derive $\nabla \cdot \mathbf{g}(\mathbf{x})$ and $\|\mathbf{g}(\mathbf{x})\|^2$ in closed form.

(b) **(14)** Apply SURE to obtain

$$\hat{R}^\lambda(\mathbf{x}) \;=\; p \;-\; 2 \cdot \#\{i : |x_i| \leq \lambda\} \;+\; \sum_{i=1}^{p} \min(x_i^2,\, \lambda^2).$$

(c) **(20)** Take $\theta = \mathbf{0}$, so each $X_i \sim \mathcal{N}(0,1)$ iid. Using the unbiasedness of SURE, derive the closed form

$$R(\mathbf{0}, \delta^\lambda) \;=\; 2p\big[(1 - \Phi(\lambda))(1 + \lambda^2) - \lambda\phi(\lambda)\big],$$

where Φ, ϕ are the standard Normal cdf and pdf. (Hint: $\mathbb{E}[Z^2 \mathbf{1}\{|Z| \leq \lambda\}] = 2\Phi(\lambda) - 1 - 2\lambda\phi(\lambda)$ for $Z \sim \mathcal{N}(0,1)$.) Show that $R(\mathbf{0}, \delta^\lambda) \to 0$ as $\lambda \to \infty$.